

Software Engineering LAB - 01

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PROBLEM STATEMENT:

Academic Elective Bidding and Allocation System

REQUIREMENTS TABLE

FR-001		Functional	The system shall allow students to distribute 100 bidding credits across their ranked elective preferences and validate prerequisite course completions.	High	Pass: Credits deduct accurately and prerequisites validated. Fail: Total allocated credits exceed 100 or unmet prerequisites bypassed.	Core bidding mechanism; ensures fair credit-based ranking and academic eligibility. (Given)
FR-002 (Neha Rastogi)		Functional	The system shall allow the Academic Registrar to configure the elective course catalog, including course list, prerequisites,	High	Pass: All configured courses appear correctly in the bidding interface with accurate seats/slots. Fail: Any course missing capacity or slot data.	Bidding and allocation cannot begin without a correctly configured catalog.

			seat capacity, and timetable slots for the term.			
FR-003		Functional	The system shall allow the Academic Registrar to execute the allocation solver, which assigns electives to students based on submitted bids, seat capacity, and prerequisite eligibility while eliminating timetable clashes.	High	Pass: Output allocation has zero students with overlapping-slot electives and zero courses exceeding seat capacity. Fail: Any conflict or overcapacity found in the output.	Core engine functionality that fulfils the fair, constraint-respecting allocation promised to students.
FR-004		Functional	The system shall allow students to view their final elective allocation and resulting timetable once the allocation solver has completed a run.	High	Pass: Allocated course(s) and schedule display accurately within 2 seconds of solver completion. Fail: Result missing, delayed, or shows unallocated/incorrect data.	Students need timely, accurate visibility into their outcome to plan their semester.
FR-005		Functional	The system shall allow students to withdraw or modify a submitted bid at any time before the bidding window closes.	Medium	Pass: Updated bid reflects the new credit distribution and previous bid is fully overwritten. Fail: Edit accepted after deadline or partial update applied.	Gives students flexibility to correct mistakes without registrar intervention, up to a firm cut-off.
NFR-001		Nonfunctional	The final elective allocation solver shall process 5,000 student bids and resolve schedule conflicts in under 30 seconds.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.	Ensures the allocation engine scales to full-university peak load without delaying result publication. (Given)
NFR-002		Nonfunctional	The system shall log every allocation solver run (timestamp, registrar ID,	Medium	Pass: Unauthorized (Student) execution attempts are rejected;	Supports academic-integrity audits and dispute

			input bid snapshot, output allocation) and restrict solver execution to users with Registrar privileges.		every run produces an immutable audit record. Fail: Solver runs without logging or is triggerable by a non-Registrar account.	resolution when students appeal their allocation.
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USE-CASE FLOW:

Actor

Student (primary actor). The Academic Registrar and the Allocation Solver Engine are indirectly involved, since this use case only produces results after they complete UC-02 (Configure Elective Courses) and UC-03 (Execute Allocation Solver).

Preconditions

1. The student is authenticated and has an active account for the current academic term.
2. The student has submitted a valid elective bid (UC-01: Submit Elective Bid).
3. The Academic Registrar has successfully executed the allocation solver (UC-03) for the current term, and a result set exists.

Postconditions

1. The student has viewed their confirmed course, timetable slot, and credits spent for the current term.
2. The system has recorded an audit-log entry for the view event (NFR-002: timestamp + user ID).

Main Success Scenario

1. Student logs into the elective bidding portal and selects “View Final Allocation.”
2. System checks whether a completed allocation run exists for the student for the current term.
3. System retrieves the student's allocated course(s) — course name, timetable slot, and credits spent — from the solver's output.
4. System displays the allocation clearly (course code, section, day/time, room, credits used).
5. System logs the view event (student ID, timestamp) for audit purposes.
6. Use case ends successfully.

Alternate Flow A — Allocation Not Yet Published (extends step 2)

- 2a1. If the allocation solver has not yet run for the current term, the system displays: “Allocation not yet published. Please check back after <date>.”
- 2a2. Use case ends; student may return later.

Alternate Flow B — Student Disputes the Allocation («extend» by UC-05 Appeal Allocation, from step 4)

4a1. Student selects “Appeal this allocation” from the results screen.

4a2. System launches the Appeal Allocation use case (FR-005), accepting the appeal only if within the 48-hour publication window.

4a3. Use case resumes or ends according to the Appeal Allocation flow.

